[5] We'll start using the notation $\vec{\sigma}$, except when referring to components, such as σ_x.

[6] The careful reader may object, because the result of this "ordinary" dot product is a 2 × 2 matrix rather than a scalar, so it's not *quite* ordinary. Perhaps there is some comfort in the fact that the resulting matrix operator corresponds to a vector component, which *is* a scalar. It all works out in the end.